Han Liu, Ph.D.

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SUMMARY

I write in-house Computational Fluid Dynamics (CFD) solvers from governing equations to production code: numerical schemes, parallel computing (MPI, CUDA), CPU/GPU optimization. My work focuses on multiphase turbulence across incompressible and compressible regimes, with applications in ocean engineering, environmental response, public health, and high-energy physics. Now pushing CFD into its next generation by combining quantum computing, AI, and High-Performance Computing (HPC). Also consulting on physics-based AI foundational models and digital twin platforms.

EDUCATION

Peking University 2012 – 2018
Ph.D. in Mechanical Engineering, Fluid Mechanics Beijing, China

- Thesis: Characteristics and action mechanism of the structures in ventilated supercavity

Peking University 2008 – 2012
B.S. in Theoretical and Applied Mechanics Beijing, China

- Thesis: Numerical investigation on heavy particle pair dispersion in the inertial range of two dimensional turbulence

IBM Qiskit Global Summer School on Quantum Machine Learning 2021
Certificate of Quantum Excellence in Quantum Computing and Quantum Machine Learning

- Grade: 100/100

CONSULTING EXPERIENCE

VINCI4D.ai Mar 2026 – Jun 2026
Consultant, Foundational AI Model

- Leveraged AI agentic workflows to develop a scalable, GPU-parallel Computational Fluid Dynamics (CFD) solver with adaptive mesh refinement (AMR) for chip-level thermal analysis.
- Engineered and optimized custom GPU kernels for high-throughput, large-scale parallel thermal simulation.
- Generated high-fidelity training data underpinning a physics-based foundational AI model for semiconductor chip-design customers.

Claros Technologies Jan 2025 – Present
Consultant, Digital Twin Platform

- Built end-to-end CFD digital twin for PFAS chemical reactor with parametric geometry, simulation, data generation, and optimization.
- Integrated chemical kinetics into the solver and developed optimization algorithms for reactive flow design.

DDT (Dynamic Dimension Technologies, LLC) Jan 2023 – Sep 2023
Consultant, Ocean Wave Digital Twin

- Built ocean wave modeling platform (shoaling, bathymetry, vehicle dynamics) with synthetic data generation for surrogate models.

ACADEMIC EXPERIENCE

GHD-Sponsored Research, St. Anthony Falls Laboratory, University of Minnesota Oct 2026 – Present
Lead PI, PFAS–Microplastic Interactions (with Dr. Zhiyong Xia, GHD) Minneapolis, MN

- Secured and lead an industry-sponsored project for GHD quantifying how seven PFAS (PFOA, PFOS, PFBA, PFBS, TFA, TFMS, 6:2 FTS) bind to PE and PET microplastics, which interactions control the binding, and how weathering (C=O, OH, COOH, COO^- groups) and pH change it.

- Built a quantum-chemistry workflow: CREST conformer searches, dispersion-corrected DFT (ωB97M-V/def2-TZVPD with SMD solvation and counterpoise correction) for 22 PFAS–polymer complexes with frequency-verified minima, and SAPT energy decomposition.
- Mapped surface electrostatic potentials of PE, PET, PVC and PS; benchmarked the functional against CCSD(T); ran GPU-accelerated quantum chemistry (gpu4pyscf) on A100/H100 clusters.

St. Anthony Falls Laboratory, University of Minnesota Jan 2021 – Present
Researcher Minneapolis, MN

- Built an incompressible CFD solver with AMR, coupled level-set/VOF interface tracking, and cut-cell embedded boundaries on AMReX (26K lines of C++, 12 modules, 467 unit tests, 7 validated benchmarks).
- Ported lab's Fortran HPC code to C++/CUDA multi-GPU. Breaking wave case ($2048\times2048\times1024$): 1 month on 2048 CPUs $\rightarrow$ 2.5 days on 64 A100s $\rightarrow$ under 5 hours with mixed-precision Poisson solver. 40% memory reduction via dynamic memory allocation and custom memory pool.
- Developed hybrid quantum-classical framework (variational quantum circuits + physics-informed neural networks) for incompressible turbulence simulation.
- Applied quantum tensor networks (MPS/MPO) for data compression and acceleration in Navier-Stokes simulations.

St. Anthony Falls Laboratory, University of Minnesota Sep 2018 – Jan 2021
Post-Doctoral Associate Minneapolis, MN

- Developed coupled level-set/VOF interface tracking for two-phase and three-phase flows at high density ratios ($>$1000:1); applied to oil spill simulation in turbulent breaking waves (JCP 2023).
- Simulated COVID-19 airborne transmission in indoor environments; **Featured Article** in Physics of Fluids, covered by The Guardian.
- Researched wind-wave-bathymetry interactions near the surf zone for amphibious vehicle performance.
- Optimized hydro-turbine geometries using immersed boundary methods and large-eddy simulation.

University of Minnesota Sep 2015 – Aug 2017
Visiting Scholar

- Developed in-house code for simulating multiphase turbulent flow with VOF, level set, and immersed boundary methods.
- Researched turbulent cavitating flows interacting with ocean waves.

Peking University Sep 2012 – Jun 2018
Research Assistant Beijing, China

- Built a compressible high-Mach-number multiphase solver from scratch (Fortran + MPI); applied to shock–multiphase-flow interactions for inertial confinement fusion (Physical Review E).
- Developed coupled level-set/VOF interface tracking in an incompressible flow solver; studied underwater cavitation drag reduction (Journal of Fluid Mechanics).
- Investigated Richtmyer-Meshkov instability and turbulent mixing.

GRANTS

GHD — Computational Investigation of PFAS–Microplastic Interactions Nov 2026 – Feb 2027
Lead PI (Co-PI: Lian Shen); industry-sponsored research, $10,000

RESEARCH INTERESTS

Computational Fluid Dynamics (CFD): High-fidelity numerical simulation (DNS/LES), level set and volume of fluid methods, adaptive mesh refinement (AMR), immersed boundary and cut-cell methods.

Multiphase Flow: Air-water-oil systems, ventilated cavitation, bubble dynamics and tracking, high-density-ratio interface tracking.

Turbulence: Compressible turbulence, turbulent mixing, Richtmyer-Meshkov instability.

High-Performance Computing: Multi-GPU parallel computing (C++/CUDA), mixed-precision solvers, large-scale simulation optimization.

Applied CFD: COVID-19 airborne transmission, oil spill dynamics, hydrokinetic turbine design, aquatic canopy flows, digital twin platforms.

Quantum Computing: Tensor network methods for Navier-Stokes equations, variational quantum algorithms for fluid mechanics, matrix product state encoding.

SKILLS

HPC: Fortran, C++, CUDA, MPI, Python, Git, Slurm, PBS, HDF5, PETSc, hypre.

Multiphase CFD: High-order schemes, interface tracking (level set, VOF, CLSVOF), Basilisk, AMReX, DNS/LES/RANS, turbulence modeling, immersed boundary method, shock capturing, Poisson solver, fluid-structure interaction, finite difference/volume, OpenFOAM, ANSYS.

AI/ML: Physics-informed neural networks (PINNs), LLM-augmented development, multi-agent orchestration.

Quantum Computing: Qiskit, Yao.jl, variational quantum circuits, quantum-PINNs, quantum tensor networks.

SELECTED PUBLICATIONS

- **Liu, H.**, Li, T., Shen, L. "Quantum-inspired tensor network representations of the shallow water equation." *Under revision*.
- Basak, S., Bitencourt, U.C., Gembali, S.C., **Liu, H.**, Shen, L. "A conservative integral formulation of surface tension in three dimensions." *Journal of Computational Physics*, 569, 115469, 2027.
- **Liu, H.**, Xiao, Z., Shen, L. "Simulation-based study of low Reynolds number flow around a ventilated cavity." *Journal of Fluid Mechanics*, 966, A20, 2023.
- Zeng, Y., **Liu, H.**, et al. "A consistent adaptive level set framework for incompressible two-phase flows with high density ratios and high Reynolds numbers." *Journal of Computational Physics*, 478, 111971, 2023.
- He, S., **Liu, H.**, Shen, L. "Simulation-based study of turbulent aquatic canopy flows with flexible stems." *Journal of Fluid Mechanics*, 947, A33, 2022.
- Gao, J., **Liu, H.**, Lee, J., Zheng, Y., Guala, M., Shen, L. "Large-eddy simulation and co-design strategy for a drag-type vertical axis hydrokinetic turbine." *Renewable Energy*, 181, 1305–1316, 2022.
- Wu, J., **Liu, H.**, Xiao, Z. "Refined modeling for single-mode cylindrical Richtmyer-Meshkov instability." *Journal of Fluid Mechanics*, 908, 2021.
- Gao, Q., Deane, G.B., **Liu, H.**, Shen, L. "A robust and accurate technique for Lagrangian tracking of bubbles and detecting fragmentation and coalescence." *International Journal of Multiphase Flow*, 135, 103523, 2021.
- Wang, Z., **Liu, H.**, Gao, Q., et al. "Numerical investigation of ventilated cavitating flow in the wake of a circular cylinder." *Physical Review Fluids*, 6, 064303, 2021.
- **Liu, H.**, He, S., Shen, L., Hong, J. "Simulation-based study of COVID-19 outbreak associated with air-conditioning in a restaurant [**Featured Article**]." *Physics of Fluids*, 33(2), 023301, 2021.
- Wu, J., **Liu, H.**, Xiao, Z. "Numerical investigation of Richtmyer-Meshkov instability in spherical geometry." *Advances in Applied Mathematics and Mechanics*, 11(3), 583–597, 2019.
- **Liu, H.**, Xiao, Z. "Scale-to-scale energy transfer in mixing flow induced by the Richtmyer-Meshkov instability." *Physical Review E*, 93, 053112, 2016.
- Zhang, Q., **Liu, H.**, Ma, Z., Xiao, Z. "Preferential concentration of heavy particles in compressible isotropic turbulence." *Physics of Fluids*, 28, 055104, 2016.

SELECTED PRESENTATIONS

"Enhancing incompressible two-phase turbulence simulations with quantum tensor networks" 2025
APS Global Summit 2025, Anaheim, CA

"Quantum-inspired high-fidelity solver for simulating incompressible two-phase turbulence" 2024
77th APS Division of Fluid Dynamics Annual Meeting, Salt Lake City

"Efficiency-enhanced high-fidelity simulation solver for incompressible two-phase flows and fluid-structure interactions" 2023
76th APS DFD Annual Meeting

"Simulation-based study on turbulent canopy flows and Monami phenomenon" 2022
75th APS DFD Annual Meeting, Indianapolis

"Simulation-based study on the interaction between flexible canopy and turbulent boundary layer flows" 2022
AGU 2022, Chicago

"Variational quantum algorithm towards quantum computing for fluid mechanics" 2021
74th APS DFD Annual Meeting, Phoenix

"Simulation-based study of airborne transmission of COVID-19 in two practical settings" 2020
73rd APS DFD Annual Meeting (Virtual)

"Simulation-based study of wind effect on shoaling waves" 2020
Ocean Sciences Meeting, San Diego
"A stabilized coupled level set and volume of fluid method" 2019
72nd APS DFD Annual Meeting, Seattle
"Numerical study on supercavitation turbulence" 2018
71st APS DFD Annual Meeting, Atlanta
"Simulation based study of supercavitation turbulence" 2017
10th International Workshop on Ship and Marine Hydrodynamics, Taiwan
"Numerical study on the interaction between supercavitation and turbulence" 2016
69th APS DFD Annual Meeting, Portland
"Preferential concentration of particles in compressible turbulence" 2015
15th European Turbulence Conference (ETC15), Delft, Netherlands
"Direct numerical simulation of the turbulent mixing in Richtmyer-Meshkov Instability" 2014
6th European Conference on Computational Fluid Dynamics (ECFD VI), Barcelona
"Energy transfer during the turbulent mixing in Richtmyer-Meshkov Instability" 2014
2nd International Conference on High Energy Density Physics (ICHEDP), Beijing

PEER REVIEW EXPERIENCE

Physical Review Letters, Journal of Fluid Mechanics, Geophysical Research Letters, Physics of Fluids, Journal of Computational Physics

SERVICE

Session Chair: Computational Fluid Dynamics: General III, APS Division of Fluid Dynamics Annual Meeting 2021

MEDIA COVERAGE

- "Simulation-based study of COVID-19 outbreak associated with air-conditioning in a restaurant" (*Physics of Fluids*, 2021) – covered by The Guardian.